



Wing Flexibility Induced Control Reversal For Flapping Wing Vehicles: Theoretical Analysis

David O. Sigthorsson ^{*}
Michael W. Oppenheimer [†]
David B. Doman [‡]
Isaac E. Weintraub [§]

Control reversal due to wing flexibility is of concern when designing maneuverable aircrafts. Experiments with prototype minimally actuated flapping wing micro air vehicles using split-cycle constant period frequency modulation for control revealed a control effect reversal when using flexible wings to produce desired wing rotations. When controlling the cycle-averaged forward translational force and rolling moment using split-cycle constant period frequency modulation; as wing stroke symmetric frequency is increased, the effect of a given split-cycle command reverses. This study investigates how a simple spring model augmentation of a quasi-steady aerodynamics model with a rigid body vehicle can account for the observed reversal as the dynamic pressure on the wing becomes coupled with the angle-of-attack. Furthermore, the feasibility of employing split-cycle constant period frequency modulation for control around a practical hover condition is established despite the effect reversal. A companion paper investigates the same phenomenon using observations and empirical analysis.

I. Introduction

Flapping wing micro air vehicles (FWMAV) have been of particular research interest for the past decade due to the impressive flight capabilities of insects and small birds, and their potential application for military and civilian surveillance, remote sensing, low impact infiltration, and entertainment. Several researchers have shown success or great promise in developing FWMAVs.¹⁻⁴ The work in this study focuses on minimally actuated FWMAVs utilizing split-cycle constant-period (SCCP) frequency wingbeat modulation to implement controlled flight around hover with only two actuators.^{5,6} In theoretical works and early prototypes, a fixed angle-of-attack (AOA) was enforced on each wing over each half stroke^{4,5} which enables SCCP modulation to control a cycle-averaged translational force and the rolling moment. Recent works relied on dynamic pressure and wing flexibility to produce passive wing rotation resulting in a desirable AOA distribution over the span

^{*}Research Scientist IV, InfoSciTex DCS Corp, 2210 Eighth Street, Bldg. 146, Rm. 301, Air Force Research Laboratory, WPAFB, OH 45433-7531 Email David.Sigthorsson.ctr.is@us.af.mil, Fax (937) 656-4000, Member AIAA

[†]Senior Electronics Engineer, Control Science Center, 2210 Eighth Street, Bldg. 146, Rm. 301, Air Force Research Laboratory, WPAFB, OH 45433-7531 Email Michael.Oppenheimer@us.af.mil, Fax (937) 656-4000, Associate Fellow AIAA

[‡]Principal Aerospace Engineer, Control Science Center, 2210 Eighth Street, Bldg. 146, Rm. 301, Air Force Research Laboratory, WPAFB, OH 45433-7531 Email David.Doman@us.af.mil, Fax (937) 656-4000, Fellow AIAA

[§]Electronics Engineer, Aerospace Vehicle Technology Assessment And Simulation Branch, 2210 Eighth Street, Bldg. 145, Rm. 40, Air Force Research Laboratory, WPAFB, OH 45433-7531 Email IsaacWeintraub.1@us.af.mil, Fax (937) 656-4000, Member AIAA

of each wing that facilitated the production of lift and augmented drag forces. With appropriate selection of wing structural and aerodynamic characteristics, along with flapping frequency, the prototype vehicle was capable of providing greater lift than weight;⁷ however, the force and moment sensitivity to SCCP modulation was observed to be the reverse of those observed at lower flapping frequencies and those predicted by the fixed AOA models and trials.^{4,5} The objective of this paper and its companion paper⁸ is to analyze and account for this apparent reversal of control effectiveness arising from wing flexibility. In particular, investigating if the quasi-steady aerodynamics model used in previous force and moment analyses is capable of accounting for the observed effect under a simple first principles based wing flexibility model, coupling the dynamic pressure with the angle-of-attack. The companion paper⁸ addresses the same question using observations, measurements, and analyses.

Control reversal under increasing dynamic pressure due to wing flexibility is a known phenomenon in aerospace practice and literature.^{9,10} Conventionally, the reversal is to be avoided or seen as a design fault, requiring either increasingly restricted control deflections under increasing dynamic pressure, or increasing the wing stiffness to avoid the control reversal entirely in a given operating range. These responses are not applicable for the class of flexible flapping wings considered here. Firstly, wing flexibility is essential to provide a practical AOA operating range and to attain hovering flight. Without flexibility, additional mechanisms would be needed to provide wing rotation, which is counter to the design objective of minimizing both structure and actuation. Thus simply stiffening the wing is not a viable solution to avoid control reversal. Secondly, it will be shown that the reversal primarily affects the cycle-averaged drag, while minimally impacting the cycle-averaged lift. Thus, operating at relatively high dynamic pressures may be viable in that: adequate lift is still provided, while cycle-averaged drag manipulation by SCCP modulation of the wingbeat provides sufficient control effectiveness with translational force and rotational moment sensitivities being reversed from expected results at low dynamic pressures.

The coupling between the dynamic pressure and AOA is analyzed using a simplified first principles model. The vehicle is a rigid body with two stiff wings that are each mounted at a shoulder joint allowing rotation about the spar limited by a hypothetical circular spring anchored to the wing center-of-pressure and a point on the stroke plane that moves with the spar. Given this configuration, an equilibrium condition is computed, relating any fixed dynamic pressure on the wing to a corresponding AOA such that the aerodynamic and spring forces balance. In order to perform tractable analytic analyses, the model is simplified to ignore the spring dynamics but stipulate that the equilibrium condition holds for any given wing velocity. The wing stroke angle is a triangular waveform, so the wing stroke velocity is fixed over each half stroke, thus the dynamic pressure and AOA remain constant over each half stroke. The sensitivities of the cycle-averaged forces and moments are then evaluated according to changes in dynamic pressure and SCCP modulation.

II. Modeling

The vehicle model is presented in Figure 1 and follows previous works on SCCP with the only notable exception being that the wing rotation about the spar is limited by a spring rather than being enforced by passive wing rotation and a stopper.⁵ Major assumptions include

- Rigid body model.
- Rigid wing.
- Wing rotation about the spar is limited by a simplified spring model.
- Quiescent air and no wing-to-wing coupling.

- Quasi-steady aerodynamics model (same or similar to the model by Sane and Dickinson¹¹).

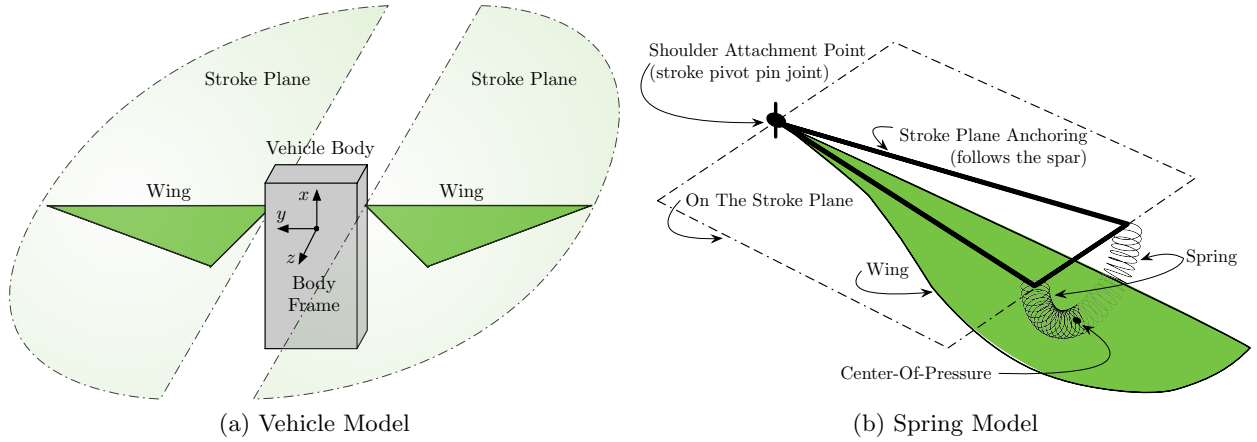


Figure 1: Conceptual vehicle and spring model.

The wing is free to rotate about its shoulder in the stroke plane which is orthogonal to the vertical axis (x -axis) of the vehicle, with 0° being straight out to the side. The wing stroke angle, ϕ , is a triangular waveform. Over each upstroke, $t \in [0, T_u]$,

$$\phi(t) = \phi_u(t) = t\dot{\phi}_u + A \quad , \quad \dot{\phi}_u = -\frac{2A}{T_u} \quad (1)$$

where A is the amplitude of the stroke and T_u is the time duration of the upstroke. Similarly, by shifting the time variable so $t \in [0, T_d]$, over each downstroke the stroke angle is

$$\phi(t) = \phi_d(t) = t\dot{\phi}_d - A \quad , \quad \dot{\phi}_d = \frac{2A}{T_d} \quad (2)$$

where T_d is the duration of the downstroke. The duration of a full wingstroke is clearly

$$T = T_u + T_d \quad (3)$$

Using the SCCP notation, the split-cycle symmetric frequency and split-cycle parameter are

$$\omega = \frac{2\pi}{T} \quad , \quad \delta = \omega - \frac{\pi}{T_u} \quad (4)$$

respectively.

The aerodynamic forces are the lift and drag acting on the center-of-pressure of the wing

$$\begin{aligned} L &= \frac{\rho}{2} I_A \dot{\phi}^2 C_L(\alpha) \\ D &= \frac{\rho}{2} I_A \dot{\phi}^2 C_D(\alpha) \end{aligned} \quad (5)$$

where ρ is the air density, I_A is the second area moment of the wing, α is the angle of attack and the lift and drag coefficients are

$$\begin{aligned} C_L(\alpha) &= 1.58 \sin(2.13\alpha - 0.1257) + 0.22 \\ C_D(\alpha) &= -1.55 \cos(2.04\alpha - 0.1714) + 1.92 \end{aligned} \quad (6)$$

respectively, from Sane and Dickinson.¹¹ For analytic analysis, the lift and drag coefficients are approximated as

$$\begin{aligned} C_L(\alpha) &= c_a \sin(2\alpha) &= 2c_a \cos(\alpha) \sin(\alpha) \\ C_D(\alpha) &= c_a(1 - \cos(2\alpha)) = 2c_a \sin(\alpha)^2 \end{aligned} \quad (7)$$

for some constant c_a . In particular, $c_a = 1.7628$ is a least-mean-squares fit to the Sane and Dickinson coefficients. The total aerodynamic force acting on the center-of-pressure of the wing and against the wing motion is

$$F_{\text{aero}} = L \cos(\alpha) + D \sin(\alpha) \quad (8)$$

Using substitution from (7) and elementary algebra and trigonometry yields

$$F_{\text{aero}} = \frac{\rho}{2} I_A \dot{\phi}^2 (2c_a \sin(\alpha)) \quad (9)$$

The spring force is modeled according to a linear spring, Hooke's law, acting at the center-of-pressure normal to the wing surface, against deviation of the wing rotation from $\alpha = \pi/2$,

$$F_{\text{spring}} = k_s(\pi/2 - \alpha) \quad (10)$$

where k_s is the combined spring coefficient.

The force diagram in Figure 2 shows the aerodynamic and spring force acting on the wing. The equilibrium condition where the aerodynamic force equals the spring force is described as

$$F_{\text{aero}} = F_{\text{spring}} \quad (11)$$

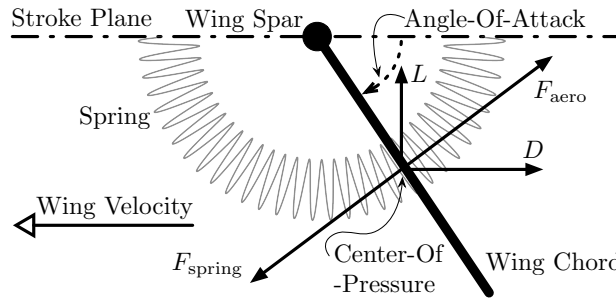


Figure 2: Spring and aerodynamic force diagram, view along wing spar.

Stipulating that equilibrium (11) holds at any given time yields the wing velocity, or dynamic pressure, as a function of the AOA, namely from (9) and (10)

$$\frac{\rho}{2} I_A \dot{\phi}^2 = \frac{k_s(\pi/2 - \alpha)}{2c_a \sin(\alpha)} \quad (12)$$

Conventionally, dynamic pressure is defined as air-density over two times velocity squared, but due to the wing flapping, the wing velocity is not uniform along the spar and hence the dynamic pressure is not uniform along the wing. The integrals involved with deriving the aerodynamic forces entangle the geometry of the wing with the dynamic pressure. Hence a “pseudo dynamic pressure” is defined as

$$\hat{q} := \frac{\rho}{2} I_A \dot{\phi}^2 \quad (13)$$

which is proportional to the dynamic pressure and with some abuse of notation will be referred to as the dynamic pressure for brevity.

The equilibrium (12) renders the following dictated relationship between the dynamic pressure and the angle-of-attack

$$\hat{q} = \frac{k_s(\pi/2 - \alpha)}{2c_a \sin(\alpha)} \quad (14)$$

The equilibrium is now described by $\hat{q}$ as a function of α , but the function can not be analytically inverted. However, the relationship yields a one-to-one and onto mapping from $\alpha \in (0, \pi/2)$ to $\hat{q} \in (0, +\infty)$, so for any given $\hat{q}$, α can be found numerically or graphically on the interval $(0, \pi/2)$. Furthermore, the sensitivity of $\hat{q}$ with respect to α can be found

$$\frac{\partial \hat{q}}{\partial \alpha} = -\frac{k_s}{2c_a} \frac{(\pi/2 - \alpha + \tan(\alpha))}{\tan(\alpha) \sin(\alpha)} \quad (15)$$

and the reciprocal yields the sensitivity of α with respect to $\hat{q}$ in closed form despite not having a closed form expression of α as a function of $\hat{q}$, namely

$$\frac{\partial \alpha}{\partial \hat{q}} = -\frac{2c_a}{k_s} \frac{\tan(\alpha) \sin(\alpha)}{(\pi/2 - \alpha + \tan(\alpha))} \quad (16)$$

Substituting the spring balance (12) into (5) the lift and drag forces become

$$\begin{aligned} L &= k_s(\pi/2 - \alpha) \cos(\alpha) \\ D &= k_s(\pi/2 - \alpha) \sin(\alpha) \end{aligned} \quad (17)$$

which is an interesting formulation of the lift and drag depending solely on the angle-of-attack. Recall that the formulation is a result of adopting reasonable approximations of the lift and drag coefficients, (7), and the angle-of-attack is dictated by the dynamic pressure from the spring equilibrium condition. The spring balance assumption needs to be kept in mind when evaluating this formulation. For example, observe that lift strictly increases with decreasing AOA, with maximum lift as $\alpha \rightarrow 0$, which seems counter-intuitive. However, recalling that $\alpha = 0$ is an unrealizable condition because it requires the dynamic pressure and stroke velocity to tend to infinity, brings the observation into line with conventional intuitions.

Figure 3 shows the aerodynamic forces, the aerodynamic coefficients, and the pseudo dynamic pressure as a function of AOA, illustrating the effect of coupling the dynamic pressure and AOA through the spring model. The lift is ever increasing as AOA decreases and dynamic pressure increases, unlike the lift curve for fixed AOA models where the lift is maximized at AOA=45° due to the concave shape of the C_L curve. Furthermore, maximal lift is now unrealizable as the dynamic pressure rises towards infinity as the AOA approaches 0°. The drag, however, does have a concave shape with a maxima at AOA=45°, diminishing to zero as AOA goes to zero but, again, that is a unrealizable condition.

III. Split-Cycle Controlled Cycle-Averaged Force and Moment Analysis

It is not feasible to control instantaneous forces and moments of flapping wing vehicles due to the periodic nature of flapping wing flight. Instead control is sought over the cycle-averaged forces and moments acting on the rigid body of the vehicle, where the relevant force or moment is time averaged over a single parameterized wing stroke period. The particular force and moment of interest are the translational force, F_z , along the z body axis and the rolling moment, M_x , around the x axis, i.e. the forward translational force and moment about the vertical axis of the

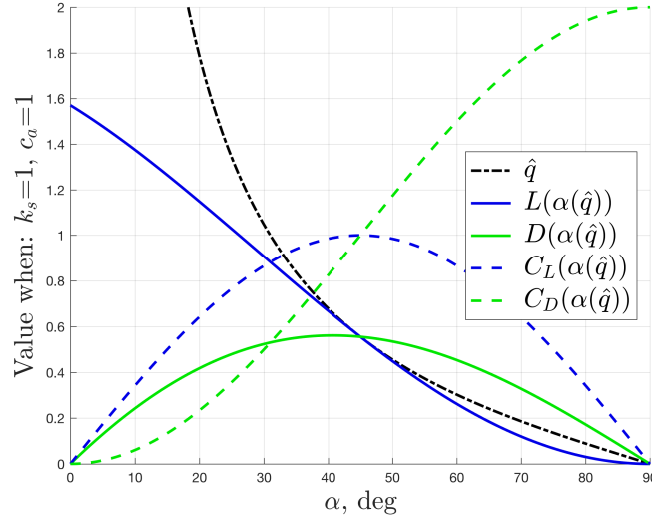


Figure 3: Aerodynamic forces, coefficients, and the pseudo dynamic pressure as functions of angle-of-attack.

vehicle during hover, respectively. These are the force and moment most strongly affected by the split-cycle parameter under SCCP wingbeat modulation⁵. By spatially symmetrically increasing the speed of the upstroke relative to the downstroke on both the left and right wing, the cycle-averaged F_z force can be varied from its nominal zero value. Similarly, by spatially asymmetrically increasing the speed of the upstroke on the right wing and downstroke on the left, the cycle-averaged M_z moment can be adjusted from its nominal zero value. The following analyses investigate the sensitivity of the cycle averaged force and moment under varying split-cycle parameters and temporally symmetric flapping frequency, by considering their sensitivity under varying dynamic pressure.

The complete derivation of the forces, moments, and their cycle-averaged expressions under SCCP modulation with a fixed AOA is given in previous works¹². Following the same approach and notation, the cycle-averaged F_z and M_x will be considered here for the right wing only since the left wing is the same with appropriate sign and notation changes.

The cycle-averaged F_z force is

$$\begin{aligned}\bar{F}_z &= \frac{1}{T} \left(\int_0^{T_u} D(\hat{q}_u) \cos(\phi_u(\tau)) d\tau - \int_0^{T_d} D(\hat{q}_d) \cos(\phi_d(\tau)) d\tau \right) \\ &= \frac{\sin(A)}{A} \frac{1}{T} \left(\hat{q}_u C_D(\alpha(\hat{q}_u)) T_u - \hat{q}_d C_D(\alpha(\hat{q}_d)) T_d \right)\end{aligned}\quad (18)$$

and the cycle-averaged M_x force is

$$\begin{aligned}\bar{M}_x &= \frac{1}{T} \left(\int_0^{T_u} D(\hat{q}_u) \left(y_{cp} + \frac{w}{2} \cos(\phi_u(\tau)) + \Delta z \sin(\phi_u(\tau)) \right) d\tau \right. \\ &\quad \left. - \int_0^{T_d} D(\hat{q}_d) \left(y_{cp} + \frac{w}{2} \cos(\phi_u(\tau)) + \Delta z \sin(\phi_u(\tau)) \right) d\tau \right) \\ &= \left(y_{cp} + \frac{w \sin(A)}{2A} + \Delta z \frac{(1 - \cos(A))}{A} \right) \frac{1}{T} \left(\hat{q}_u C_D(\alpha(\hat{q}_u)) T_u - \hat{q}_d C_D(\alpha(\hat{q}_d)) T_d \right)\end{aligned}\quad (19)$$

where subscripts u and d denote the up- and downstroke respectively, y_{cp} is the distance from the shoulder to the center-of-pressure along the spar, w is the width of the vehicle, and Δz is

the distance from the vehicle's center-of-gravity to the shoulder along the vertical axis. By using algebraic manipulation, it become clear that $\bar{M}_x$ is simply a scaling of $\bar{F}_z$ by a constant,

$$\bar{M}_x = \left(y_{cp} + \frac{w \sin(A)}{2A} + \Delta z \frac{(1 - \cos(A))}{A} \right) \frac{A}{\sin(A)} \bar{F}_z \quad (20)$$

hence analyzing $\bar{F}_z$ is sufficient to investigate both $\bar{F}_z$ and $\bar{M}_x$.

Relating $\bar{F}_z$ to only dynamic pressure and AOA will allow for analyzing the sensitivity of the force to changes in dynamic pressure. Thus, equation (18) is to be expressed in that form.

The up- and downstroke time periods are functions of $\hat{q}$, namely

$$\begin{aligned} T_u &= \frac{2A}{|\dot{\phi}_u|} = \frac{A\sqrt{k_A}}{\sqrt{\hat{q}_u}} \\ T_d &= \frac{2A}{|\dot{\phi}_d|} = \frac{A\sqrt{k_A}}{\sqrt{\hat{q}_d}} \end{aligned} \quad (21)$$

where

$$k_A := \frac{\rho}{2} I_A \quad (22)$$

Define a variable $\hat{v}$ such that

$$\hat{v}^2 := \hat{q} \quad (23)$$

The variable $\hat{v}$ will stand-in for $\hat{q}$ in our analysis as it provides a more tractable structure while maintaining a simple intuitive connection to $\hat{q}$. This structure is utilized by considering the relationship with the SCCP modulation,⁵

$$\begin{aligned} \dot{\phi}_u &= -A \frac{2(\omega - \delta)}{\pi} \\ \dot{\phi}_d &= A \frac{2(\omega + \delta \frac{\omega}{\omega - 2\delta})}{\pi} \end{aligned} \quad (24)$$

so under a small split-cycle parameter assumption,

$$\omega \gg \delta \quad (25)$$

then

$$\dot{\phi}_u = -\frac{2A}{\pi}(\omega - \delta) \quad , \quad \dot{\phi}_d \approx \frac{2A}{\pi}(\omega + \delta) \quad (26)$$

The upstroke and downstroke dynamic pressures become

$$\hat{q}_u = (\hat{v} - \Delta\hat{v})^2 \quad , \quad \hat{q}_d = (\hat{v} + \Delta\hat{v})^2 \quad (27)$$

where

$$\hat{v} = \omega \left(\frac{2A\sqrt{k_A}}{\pi} \right) = \sqrt{\hat{q}} \quad , \quad \Delta\hat{v} = \delta \left(\frac{2A\sqrt{k_A}}{\pi} \right) \quad (28)$$

and substitution into (18) with algebraic manipulation yields

$$\bar{F}_z = \frac{\sin(A)}{A} \left(\frac{\hat{v}^2 + \Delta\hat{v}^2}{2\hat{v}} \right) \left(C_D(\alpha((\hat{v} - \Delta\hat{v})^2))(\hat{v} - \Delta\hat{v}) - C_D(\alpha((\hat{v} + \Delta\hat{v})^2))(\hat{v} + \Delta\hat{v}) \right) \quad (29)$$

Using a first order Taylor approximation and a slight abuse of notation,

$$C_D(\alpha((\hat{v} \pm \Delta\hat{v})^2)) = C_D(\alpha(\hat{v}^2)) \pm \Delta\hat{v} \frac{\partial C_D(\alpha(\hat{v}^2))}{\partial \hat{v}} \quad (30)$$

Substituting 30 into 29 and applying algebraic manipulation yields

$$\bar{F}_z = -\frac{2 \sin(A)}{A} \left(\Delta\hat{v} \frac{(\hat{v}^2 + \Delta\hat{v}^2)}{2\hat{v}} \right) \left(C_D(\alpha(\hat{v}^2)) + \hat{v} \frac{\partial C_D}{\partial \hat{v}} \right) \quad (31)$$

The first two factors on the right hand side of (31) are always non-negative and scale with $\Delta\hat{v}$. Hence, the factor that dictates the sign of $\bar{F}_z$ is

$$\left(\underbrace{C_D(\alpha(\hat{v}^2))}_{\text{Direct Effect of Dynamic Pressure and Cycle-Averaging}} + \underbrace{\hat{v} \frac{\partial C_D}{\partial \hat{v}}}_{\text{"Spring Coupling"}} \right) \quad (32)$$

This factor illustrates the difference between the current model with the spring and the previous models which used a fixed AOA over each half-stroke. The first term is constant when assuming a fixed AOA and the second is zero, whereas with the spring model there can be a non-zero contribution from the second term which scales with dynamic pressure. The second term, dubbed the "spring coupling" term, arises explicitly because of the coupling of the dynamic pressure with the AOA, and thus the drag coefficient, via the spring model.

In the factor (32), C_D and factor $\hat{v}$ are always positive so $\partial C_D / \partial \hat{v}$ needs to be evaluated to investigate potential control reversal conditions,

$$\frac{\partial C_D}{\partial \hat{v}} = \frac{\partial C_D}{\partial \alpha} \frac{\partial \alpha}{\partial \hat{q}} \frac{\partial \hat{q}}{\partial \hat{v}} \quad (33)$$

Using (7), (16), and (23) yields

$$\frac{\partial C_D}{\partial \hat{v}} = \left(4c_a \sin(\alpha) \cos(\alpha) \right) \left(-\frac{2c_a}{k_s} \frac{\tan(\alpha) \sin(\alpha)}{(\pi/2 - \alpha + \tan(\alpha))} \right) (2\hat{v}) \quad (34)$$

and substitution using (23) and (14) with algebraic and trigonometric manipulation gives

$$\hat{v} \frac{\partial C_D}{\partial \hat{v}} = -8c_a \frac{(\pi/2 - \alpha) \sin(\alpha)^2}{(\pi/2 - \alpha + \tan(\alpha))} \quad (35)$$

The factor (32) becomes

$$2c \sin(\alpha)^2 - 8c_a \frac{(\pi/2 - \alpha) \sin(\alpha)^2}{(\pi/2 - \alpha + \tan(\alpha))} = \left(\frac{2c_a \sin(\alpha)^2}{(\pi/2 - \alpha + \tan(\alpha))} \right) \left(\tan(\alpha) - 3(\pi/2 - \alpha) \right) \quad (36)$$

When the factor equals zero, it defines the reversal condition at $\alpha = \alpha_r$ such that

$$\tan(\alpha_r) = 3(\pi/2 - \alpha_r) \quad (37)$$

Note that the reversal condition depends only on the AOA and has a unique solution on the interval $(0, \pi/2)$ which can be numerically found to be

$$\alpha_r \approx 58.65^\circ \quad (38)$$

Furthermore, the expression (36) consists of two factors. The first is clearly non-negative and the second has a unique zero crossing at α_r . When $\alpha \in (0, \alpha_r)$ then the expression is negative, and when $\alpha \in (\alpha_r, \pi/2)$ then the expression is positive.

The dynamic pressure determines the AOA via the spring balance, thus the reversal condition on the AOA (37) can be translated into a reversal condition on the dynamic pressure, $\hat{q} = \hat{q}_r$ where

$$\hat{q}_r = \frac{k_s(\pi/2 - \alpha_r)}{2c_a \sin(\alpha_r)} = \frac{k_s}{6c_a \cos(\alpha_r)} \quad (39)$$

by substituting (37) in (14) and simplifying. Thus,

$$\hat{q}_r \approx 0.32 \frac{k_s}{c_a} \quad (40)$$

depending only on the spring coefficient and the aerodynamic force coefficient scaling factor.

Figure 4 investigates the effects of the drag variability by graphing the terms of (32) which define the sign of $\bar{F}_z$. As described above, the sign of the term changes at $\alpha \approx 58.65^\circ$, thus defining the “reversal condition” for the control effectiveness of SSCP modulation of the wing stroke. The graph also shows how the sign of $\bar{F}_z$ agrees with fixed AOA models for higher AOA, lower dynamic pressure, and reverses as dynamic pressure is increased beyond the “reversal condition”.

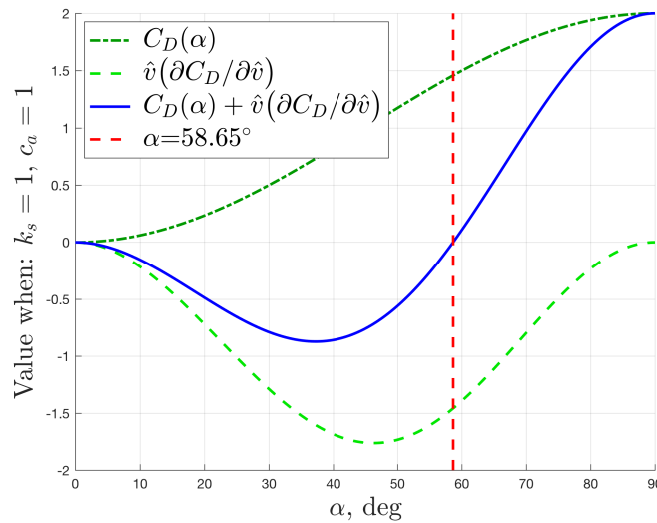


Figure 4: Term deciding the sign of $\bar{F}_z$.

The analysis above leads to the following remarks.

Remark 1 The analysis of (31) and (36) has shown that for small split-cycle parameter values such that $\omega \gg |\delta| > 0$

- For $\alpha = \alpha_r$, then $\bar{F}_z = 0$
- For $0 < \alpha < \alpha_r$, then $\bar{F}_z$ has the same sign as δ
- For $\pi/2 > \alpha > \alpha_r$, then $\bar{F}_z$ has the opposite sign of δ

The condition $\alpha = \alpha_r \approx 58.65^\circ$, defined by (37), is referred to as the “reversal condition”. The condition arises from the coupling between the dynamic pressure and the AOA via the spring model, causing the control effectiveness reversal described above.

Remark 2 The relationship between $\bar{M}_x$ and $\bar{F}_z$ described by (20), implies that Remark 1 applies to $\bar{M}_x$ the same as $\bar{F}_z$.

Remark 3 The conclusion of Remark 1 and 2 is that there is a "reversal of cycle-averaged drag" for a given small split-cycle variation as the dynamic pressure is increased under SCCP stroke modulation. In particular, control authority near symmetric flapping is lost when the AOA crosses $\alpha \approx 58.65^\circ$, satisfying (37), independent of the aerodynamic and spring coefficients, c_a and k_s . When the AOA is held above or below the "reversal condition", split-cycle parameter modulation maintains consistent control authority over the cycle-averaged forward force and rolling moment acting on the vehicle.

IV. Cycle-Averaged Lift Analysis

Control reversal on conventional aircraft, such as aileron control reversal, refer to a reversal or loss of lift due to interaction between wing flexibility and control effectors. Though it has been implied that substantial loss of lift was not the cause or expected result of this flapping wing vehicle control reversal investigation, the impact of the interplay of dynamic pressure and AOA on cycle-averaged lift is investigated for completeness.

The cycle-averaged lift is the cycle-averaged translational force along the vertical x -axis at hover, namely⁵

$$\bar{F}_x = \frac{1}{T} \left(\int_0^{T_u} L(\alpha(\hat{q}_u)) d\tau + \int_0^{T_d} L(\alpha(\hat{q}_d)) d\tau \right) \quad (41)$$

Clearly, $L(\alpha)$ is positive for all $\alpha \in (0, \pi/2)$, thus it is elementary to see that $\bar{F}_x$ is always positive. However, for completeness the analysis of $\bar{F}_x$ is carried out following the approach applied to $\bar{F}_z$ in the previous section. The cycle-averaged lift becomes

$$\begin{aligned} \bar{F}_x &= \frac{1}{T} \left(\int_0^{T_u} L(\alpha(\hat{q}_u)) d\tau + \int_0^{T_d} L(\alpha(\hat{q}_d)) d\tau \right) \\ &= \frac{1}{T} (\hat{q}_u C_L(\alpha(\hat{q}_u)) T_u + \hat{q}_d C_L(\alpha(\hat{q}_d)) T_d) \\ &\approx (\hat{v}^2 + \Delta \hat{v}^2) \left(C_L(\alpha(\hat{v}^2)) + \frac{\Delta \hat{v}^2}{\hat{v}} \frac{\partial C_L}{\partial \hat{v}} \right) \end{aligned} \quad (42)$$

The sign of $\bar{F}_x$ is clearly determined by the second factor of (42)

$$\left(\underbrace{C_L(\alpha(\hat{v}^2))}_{\text{Direct Effect of Dynamic Pressure and Cycle-Averaging}} + \underbrace{\frac{\Delta \hat{v}^2}{\hat{v}} \frac{\partial C_L}{\partial \hat{v}}}_{\text{"Spring Coupling"}} \right) \quad (43)$$

which is divided into a term from the direct effect of dynamic pressure and cycle-averaging and a "spring coupling" term. The "spring coupling" term is zero for a fixed AOA analysis but the spring model coupling the dynamic pressure and AOA yields a potentially non-zero contribution. Computing the partial derivative factor of the "spring coupling" term

$$\begin{aligned} \frac{\partial C_L}{\partial \hat{v}} &= \frac{\partial C_L}{\partial \alpha} \frac{\partial \alpha}{\partial \hat{q}} \frac{\partial \hat{q}}{\partial \hat{v}} \\ &= \left(2c_a \cos(2\alpha) \right) \left(-\frac{2c_a}{k_s} \frac{\tan(\alpha) \sin(\alpha)}{(\pi/2 - \alpha + \tan(\alpha))} \right) (2\hat{v}) \\ &= -\hat{v} \frac{8c_a^2 \tan(\alpha) \sin(\alpha) \cos(2\alpha)}{k_s (\pi/2 - \alpha + \tan(\alpha))} \end{aligned} \quad (44)$$

Then the term (43) becomes

$$c_a \sin(2\alpha) - \Delta \hat{v}^2 \frac{8c_a^2 \tan(\alpha) \sin(\alpha) \cos(2\alpha)}{k_s (\pi/2 - \alpha + \tan(\alpha))} \quad (45)$$

and by substitution with the spring balance condition (14), the term becomes

$$c_a \sin(2\alpha) - \frac{\Delta \hat{v}^2}{\hat{v}^2} \frac{4c_a \tan(\alpha) \cos(2\alpha)(\pi/2 - \alpha)}{(\pi/2 - \alpha + \tan(\alpha))} \quad (46)$$

Thus, $\bar{F}_x$ is positive when

$$\hat{v}^2 > \Delta \hat{v}^2 \frac{2 \tan(\alpha)(\pi/2 - \alpha)}{\tan(2\alpha)(\pi/2 - \alpha + \tan(\alpha))} \quad (47)$$

The factor multiplying $\Delta \hat{v}^2$ on the right hand side of (47), shown on Figure 5, depends only on α and its magnitude is less than 1 over $\alpha \in (0, \pi/2)$, thus by the assumption that the $\hat{v} \gg \Delta \hat{v}$, the inequality always holds and $\bar{F}_x$ is always positive.

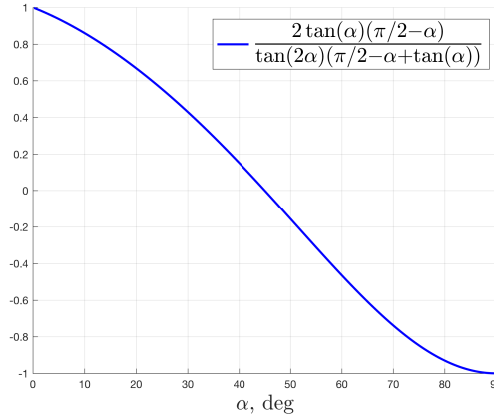


Figure 5: Analysis of the aerodynamic forces and cycle-averaged force under the spring model.

Remark 4 The cycle-averaged lift force, $\bar{F}_x$, is always positive as noted from the structure of equations (17) and (41). Thus, there is no “lift reversal” under SCCP stroke modulation.

V. Simulation

The previous sections present an analysis of the impact that coupling dynamic pressure and AOA through wing flexibility has on the sensitivities of the cycle-averaged forces and moment acting on the vehicle under small variations of the split-cycle parameter δ , i.e. using SCCP stroke modulation. In simulation, the conclusions reached in the analysis are confirmed for small values of δ and examined for larger variations in the parameter. Two parameter sets are presented as described by the two cases shown in Table 1. The academic case is simple with unity parameters, except for k_s which is set such that reversal occurs at 1 rad/sec to help illustrate the structure of the model. The practical case has parameters compatible with the vehicle presented in the companion paper based on measurement results.^{7,8}

Figure 6 shows the results of the academic case using a fixed AOA at 45° and using the spring model, drawing the contour maps of the cycle-averaged translational ($\bar{F}_z$) and lift ($\bar{F}_x$) forces. The frequency is varied from 0.5 to 1.5 rad/sec, around 1 rad/sec which corresponds to the reversal condition for small split-cycle parameters. The split-cycle parameter is varied from -0.15 to 0.15 rad/sec, or 10-30% of the frequency, which suffices to illustrate small changes in the split-cycle parameter as well as moderately large changes.^{5,8,13,14} As expected, the fixed AOA case is relatively simple. For a given split-cycle parameter, the sign of the translational force does not change as the

Parameter	Academic Case	Practical Case
Stroke Amplitude	$A = 1$	$A = 90^\circ$
Aero Coef.	$k_A = 1$	$k_A = I_A \frac{\rho}{2} = 9.65 \cdot 10^{-6} \text{kg m}$
Aero Force Coef.	$c_a = 1$	Sane and Dickinson model, equation (6)
Spring Constant	$k_s = 1.265$	$k_s = 0.3$, implies reversal at about 12Hz

Table 1: Simulation parameter table.

frequency changes, and the lift increases quadratically as the frequency increases. The spring model results show more interesting effects as frequency is varied. For a given split-cycle parameter, the sign of the translational force changes near 1 rad/sec, exactly for very small split-cycle parameters. The sign of the translational force remains consistent for a given split-cycle parameter as the frequency is varied above or below the near 1 rad/sec range. This is consistent with Remarks 1 and 3, since by design of the parameters the reversal condition of $\alpha = \alpha_r \approx 58.65^\circ$ occurs at 1 rad/sec when the split-cycle parameter is near 0.

The contour lines of the translational force in Figure 6 illustrate that as the frequency is increased or decreased by about 20% or more away from the reversal condition at 1 rad/sec, the control authority of the split-cycle over the shown range becomes consistent and significant, in that the sign of the force does not change and the magnitude is of the same order of magnitude or comparable to that obtained with the fixed AOA. For example, at flapping frequency 1.3 rad/sec, and split-cycle 0.13 and 0.026 rad/sec (about 10% and 2% variation of the frequency) the fixed AOA results show a translational force of -0.066 and -0.012 N respectively, while the spring model yields 0.048 and 0.0086 N respectively. The spring model has the opposite sign but the magnitude is over 70% of the magnitude produced with the fixed AOA case. This suggests that it is completely viable to operate in this range of higher flapping frequency and maintain a consistent sign and significant magnitude of split-cycle parameter effects.

The contour lines of the lift force in Figure 6 are quite similar for the spring model and the fixed AOA, not changing much with the split-cycle parameter, but there are some notable differences. The lift goes to lower values as the frequency is lowered for the spring model due to the AOA increasing. Similarly, the lift approaches a plateau as the frequency increases for the spring model since the AOA is reducing. However, the lift produced at a higher range of frequency is comparable in magnitude. For example, at frequencies 1.2 and 1.4 rad/sec the fixed AOA results show about 0.59 and 0.80 N respectively, while the spring model yields 0.58 and 0.79. So the lift production with the spring model is over 98% of that produced by the fixed AOA case in that range. This is not surprising since it can be read from Figure 3 that the AOA is in the neighborhood of 45° and C_L is a sinusoidal function with a maxima at 45° .

Figure 7 shows the results of the practical case based on parameters comparable to those observed in the companion paper⁸ and wing measurements paper,⁷ showing the contour lines of the cycle-averaged translational and lift forces with a fixed AOA and with the spring model. The main differences between the academic case and the practical case are that the practical case uses another set of parameters, thus expectantly causing a scaling and shifting of the results, and it uses the Sane and Dickinson lift and drag coefficient model (6) rather than the approximate (7) model. The approximate model captures the trends of the Sane and Dickinson model quite well, and hence the practical case is qualitatively very similar to the academic case, and the different values of the parameters primarily cause a quantitative scaling and shifting. The translational force reversal condition under the spring model occurs at about 12Hz similar to the academic case, so that if the frequency is increased (or decreased) by more than about 20%, the split-cycle

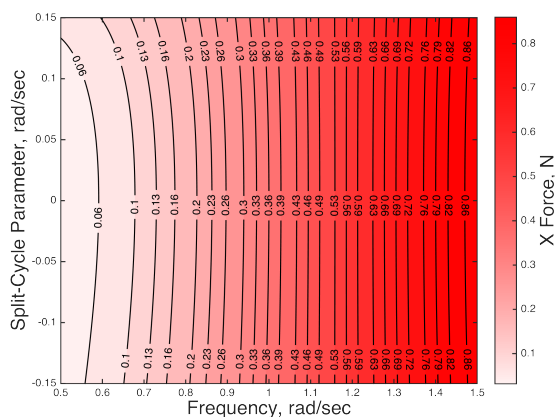
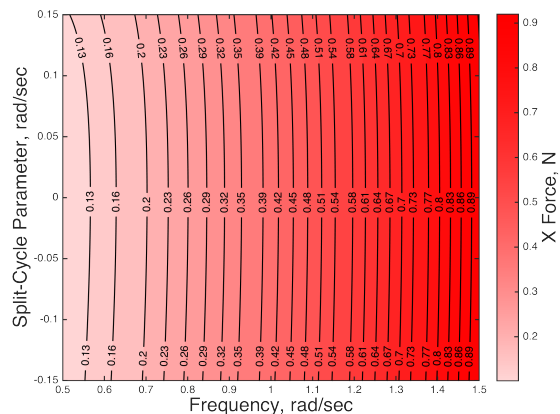
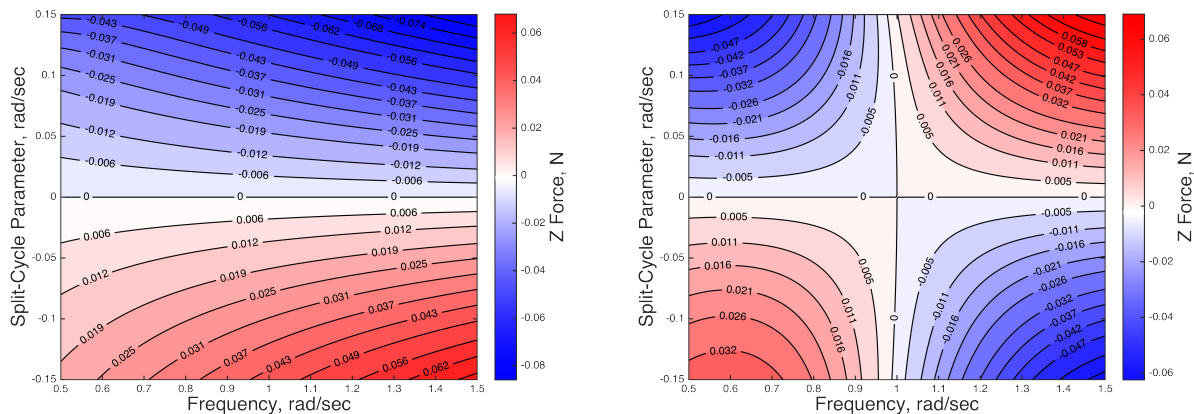


Figure 6: Academic Case Result.

parameter has a consistent and substantial affect on the translational force. In particular, from force balance measurements, it is expected that the hovering frequency is in the 18-20Hz range which is definitively in the upper range of the flapping frequency shown and sufficiently separated from the reversal condition to yield consistent and substantial control authority over the translational force when compared to the fixed AOA case.^{7,8}

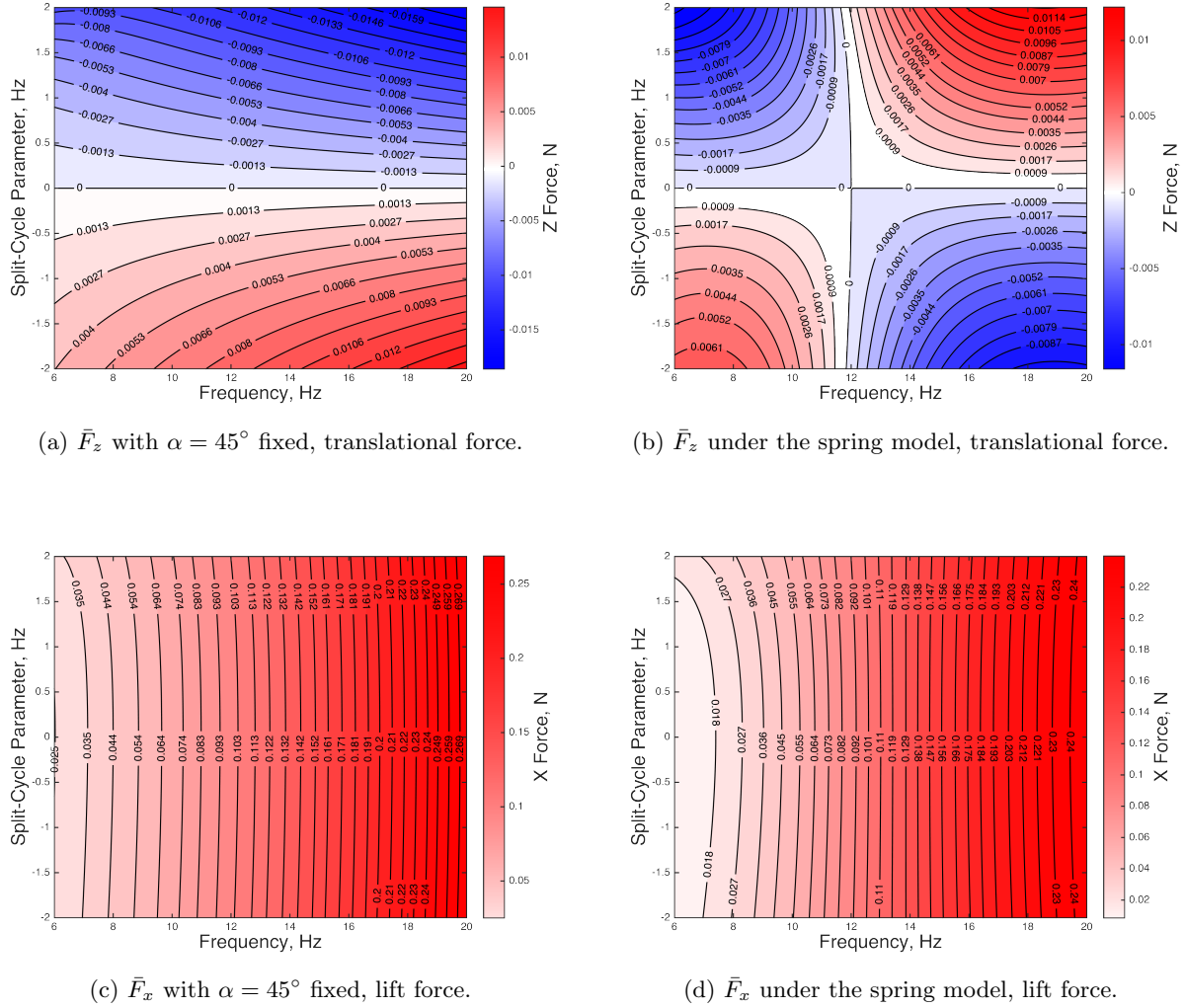


Figure 7: Practical Case Result.

VI. Conclusions

The hypothesis that the quasi-steady aerodynamics model augmented with a simplified spring model accounting for wing rotational flexibility, is sufficient to capture the combined effects of dynamic pressure and passive AOA due to torsional wing flexibility leading to control effectiveness reversal under non-zero SCCP modulation, was confirmed. The theoretical analysis with a triangular stroke waveform finds that for small deviations of the SCCP modulation, the control effectiveness reverses for the cycle-averaged translational force and rolling moment as the symmetric flapping frequency is increased. Furthermore, the reversal always occurs when the angle-of-attack is

approximately 58.65° regardless of the model parameters scaling the aerodynamic force and spring force. There is no reversal of the lift production, which strictly increases with dynamic pressure. The reason for the reversal of the cycle-averaged translational force and moment is that under low dynamic pressure, and thus a large angle-of-attack, small variations in dynamic pressure primarily cause a direct proportional impact on the drag. However, at higher dynamic pressure, and thus a smaller angle-of-attack, small variations in dynamic pressure begin to primarily indirectly impact the drag through variation of the drag coefficient due to the spring coupling between the dynamic pressure and the angle-of-attack. Both an academic and a practical parameter simulation examples confirm these conclusions and that there is a feasible operating region at a higher and lower range of symmetric flapping frequencies where SCCP modulation provides consistent and significant control authority. The higher range operating region for the practical case is likely to encompass a reasonable operating range for SCCP modulation control around hovering flight. Thus, despite the potential for control reversal in non-operating regions, SCCP modulation remains a potentially feasible control strategy for a minimally actuated flapping wing micro air vehicle.

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